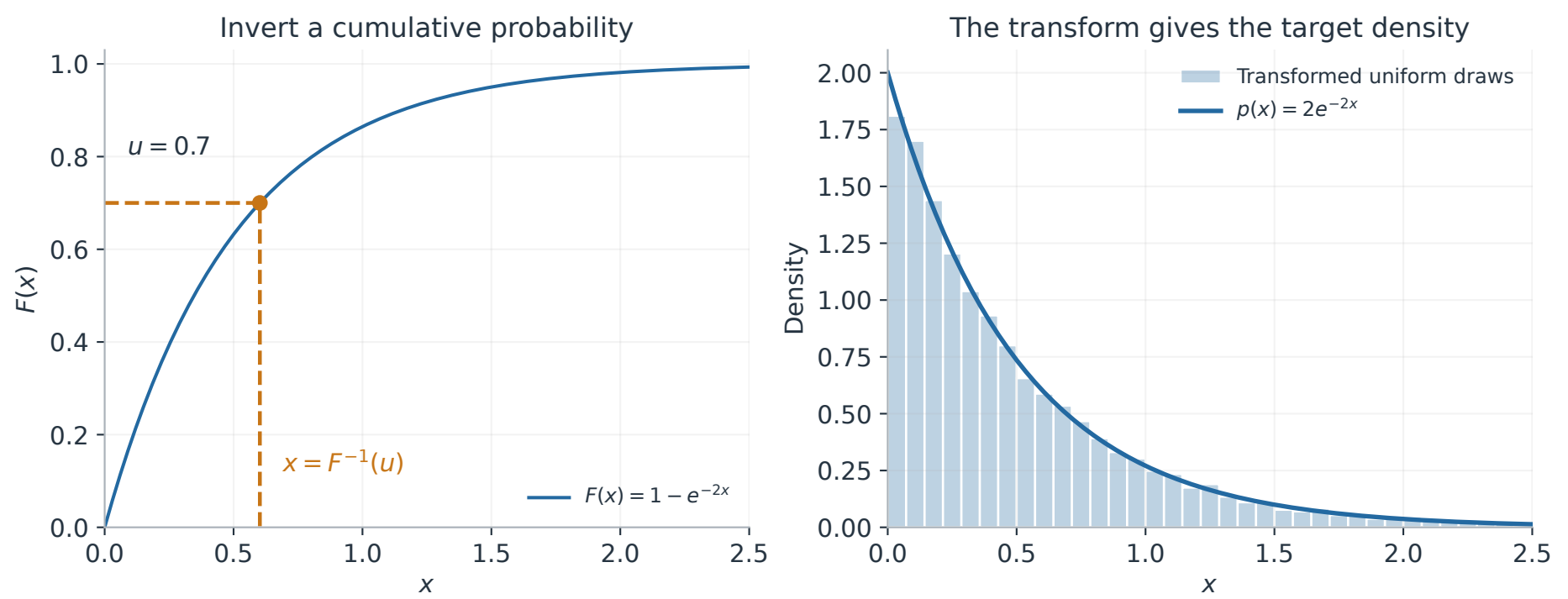
Invert a cumulative probability
$u = 0.7$
$x = F^{-1}(u)$
$F(x) = 1 - e^{-2x}$
$F(x)$
x
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
The transform gives the target density
Transformed uniform draws
$p(x) = 2e^{-2x}$
Density
x
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00